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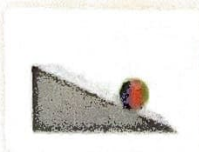
# Movements of the Earth

### Let's Learn About

- Shape of the Earth
- More about the Globe
- Movements of the Earth
- Equinoxes and Solstices
- Eclipses

### Warm Up

Things move in different ways. These movements have particular names or terms. Write the names of the movements shown with the help of the terms given in the box.



Swinging    Spinning    Rolling

### Shape of the Earth

Many years ago, people believed that the Earth was flat. But Indian astronomers Aryabhata and Bhaskara discovered that the Earth is a sphere and it spins on its axis. People did not believe them for a long time. But great explorers like Ferdinand Magellan and astronomers like Nicolaus Copernicus proved them right. Today, we know for sure that the Earth is spherical which can easily be seen in the photographs of the Earth taken with the help of satellites.

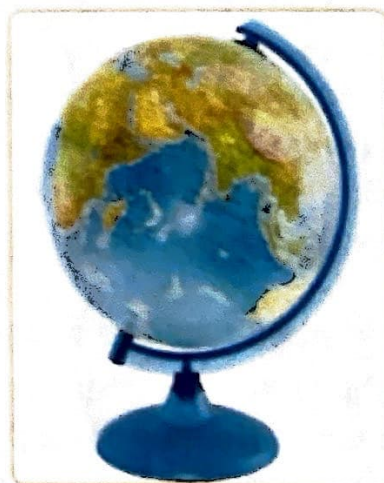


Artificial satellite taking photographs of the Earth

It is easy to learn more about the Earth with the help of a globe. (A **globe** is the model of the Earth with oceans, seas and continents shown on a smaller scale.)

### More about the Globe

Look at the globe. It spins around a rod which has two ends. This rod is not absolutely straight but is slightly tilted. In the same way, (the Earth also spins or rotates on an imaginary line called an **axis**.) The top end or the northernmost point



A globe

of the axis is known as the **North Pole** and the bottom end or the southernmost point is called the **South Pole**.

Astronomers have found that the axis of our Earth does not stand upright. It is tilted at an angle of  $23\frac{1}{2}^\circ$  from the perpendicular. This is known as the inclination of the Earth's axis.

## Movements of the Earth

The Earth has two movements. They are rotation and revolution.

### Rotation of the Earth

Rotation refers to the spinning of the Earth on its own axis. It takes approximately 24 hours to complete one rotation. The Earth rotates from west to east.

Rotation of the Earth causes day and night. When the Earth rotates on its axis, only one portion of it faces the sun and is bright. This portion of the Earth has day. The other half is away from the sun and is in darkness. This portion of the Earth has night. This occurs alternately, causing day and night, as the Earth keeps spinning.

#### Do and Learn

Take a globe. Spin it west to east because the Earth rotates in that direction. This is rotation. Now light a torch and fix it on a table. Watch where the light falls on the globe. Wherever the light of the torch falls on the globe, that part experiences daytime. The other part remains dark, which means it is night time there.

### Revolution of the Earth

Besides rotating on its axis, the Earth also moves around the sun on a fixed elliptical path known as **orbit**. This movement of the Earth is called **revolution**.

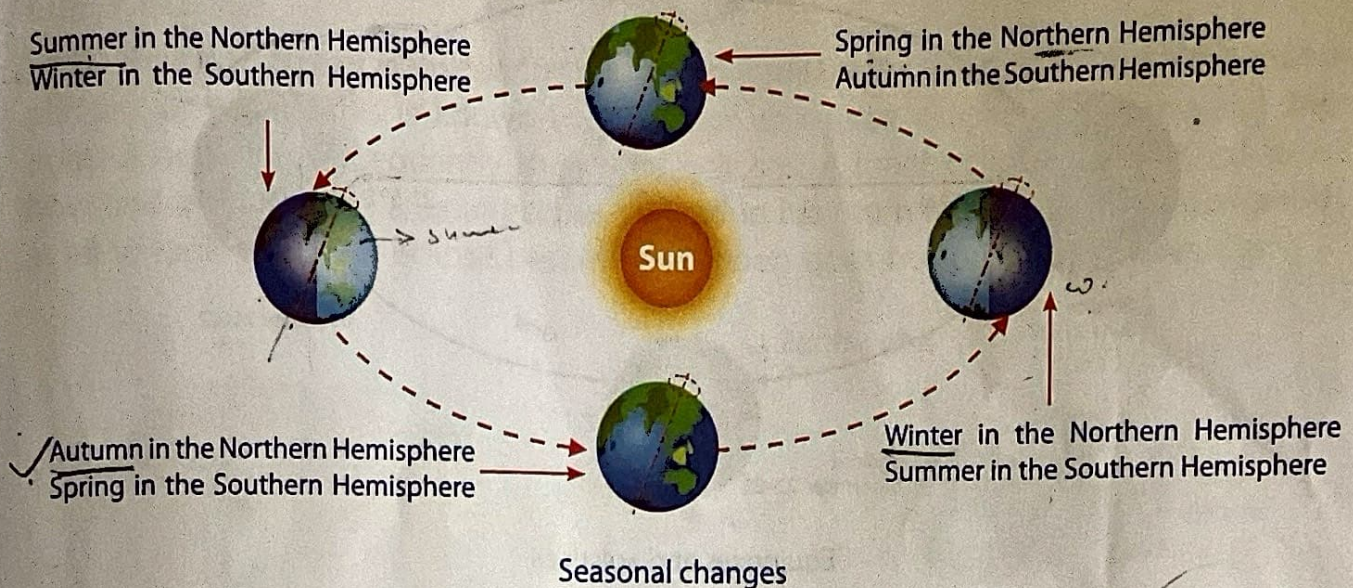
The Earth takes approximately  $365\frac{1}{4}$  days to complete one revolution. The  $\frac{1}{4}$  day is set off every year to make an extra day. This day is added to February every fourth year. This year is known as the **leap year**. Hence, every fourth year is a leap year.

As the Earth keeps rotating and revolving around the sun, different parts on the Earth receive different amount of heat and light from it. This is why the seasons keep changing. The region that is tilted away from the sun receives less heat and has winter. On the other hand, the region that is tilted towards the sun receives more heat and has summer.

### What causes the changes that we see throughout the seasons?

Throughout the year, as the Earth orbits the sun, many of its parts experience changing seasons. The warm spring brings colourful flowers. It is followed by hot summer, the

season of vacations. Then is another warm season known as autumn when the leaves turn red and brown before falling off the trees. Finally comes winter, the cold and dry season.



The tilt of the Earth's axis plays a vital role in the change of seasons. If the axis of the Earth was not tilted, it would receive the same amount of sunlight and hence the same season would be experienced throughout the year. If the Earth did not revolve around the sun, the seasons would again be the same. Different seasons are therefore caused because of the inclined axis and the revolution of the Earth around the sun.

#### Learn More

The moon takes the same period of time to complete one rotation and revolution. Hence, the same side of the moon is always seen from the Earth.

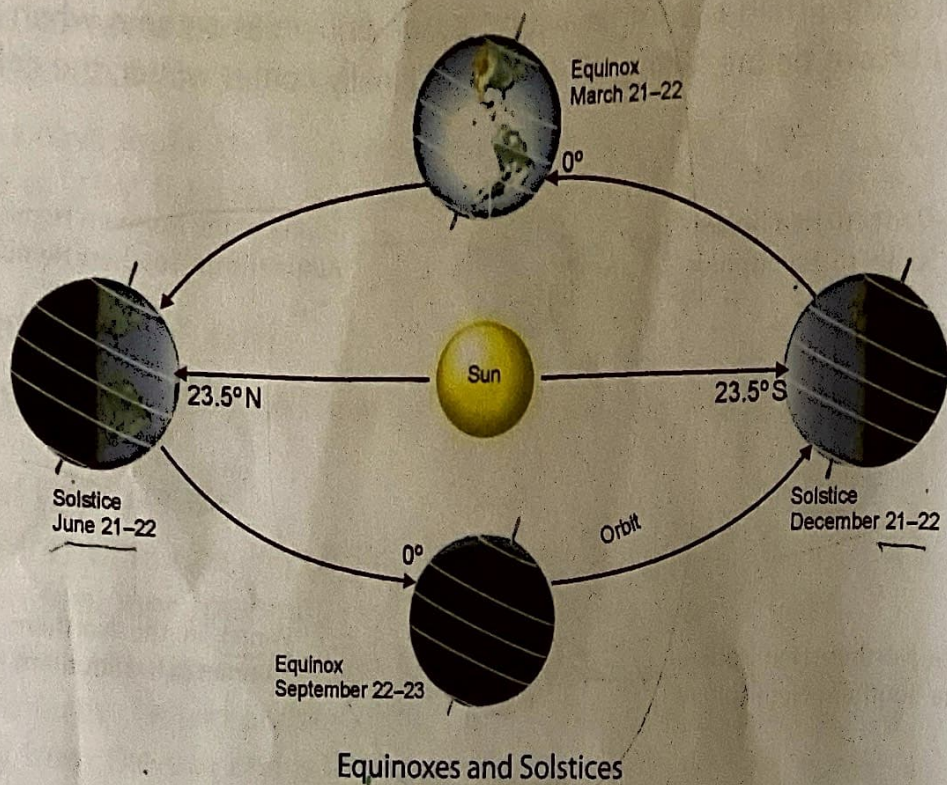
## Equinoxes and Solstices

### ✓ Equinox

21 March and 23 September are known as equinoxes when the days and nights are of equal length. During these equinoxes, the rays of the sun shine directly on the equator.

### ✓ Solstice

(21 June and 22 December are known as Summer Solstice and Winter Solstice respectively. On these solstices, the rays of the sun shine directly on one of the two tropics.) During the Summer Solstice, the sun's rays shine directly on the Tropic of Cancer. During the Winter Solstice, the sun's rays shine on the Tropic of Capricorn. The length of the days and nights is not the same during the two solstices.)

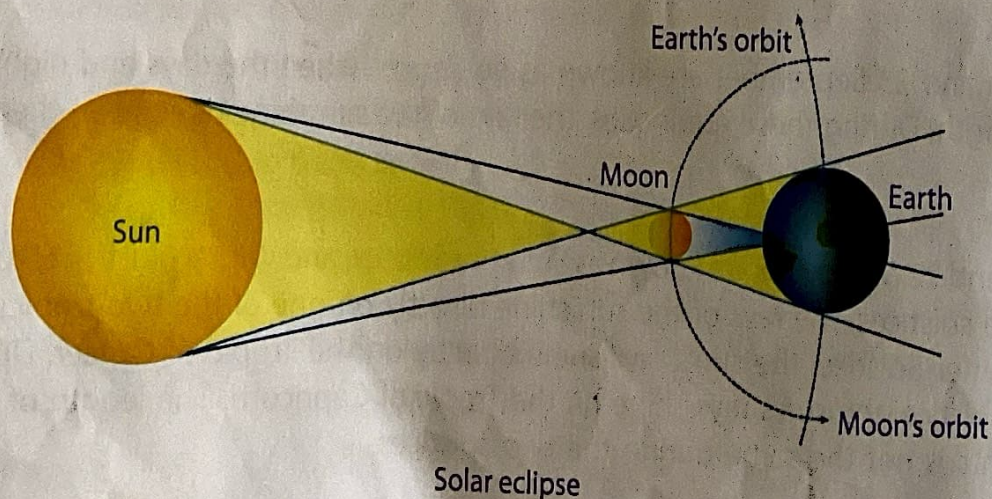


## Eclipses

The revolution of the Earth around the sun and the revolution of the moon around the Earth cause **eclipses**. This happens when the moon, the sun and the Earth fall in the same line.

### Solar Eclipse

(When the moon passes between the sun and the Earth, it blocks the rays of the sun partially or totally from falling on the Earth) It can occur only during a new moon. The Earth is then in the shadow of the moon. This is known as the **solar eclipse** or **eclipse of the sun**. It is dangerous to look at a solar eclipse directly, even if you have sunglasses on.

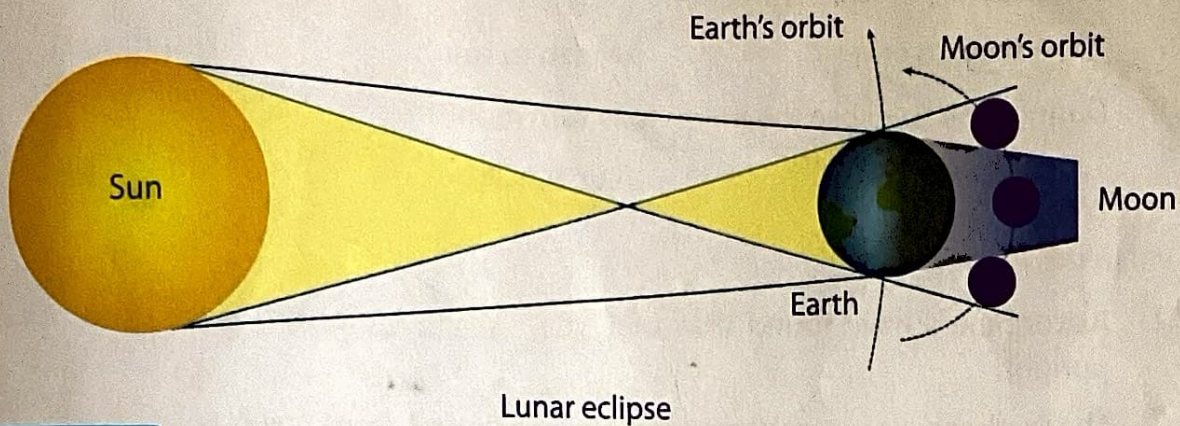


### Class Discussion

In small groups, discuss why is it dangerous to look at the solar eclipse directly.

## Lunar Eclipse

When the Earth passes between the sun and the moon, it casts a shadow on the moon, partially or totally. This is known as the **lunar eclipse** or **eclipse of the moon**. Lunar eclipses occur more frequently than solar eclipses. A lunar eclipse, can last up to an hour and a half. During a lunar eclipse, the moon may turn reddish. (It is not dangerous at all to look at a lunar eclipse because (the moon does not produce its own light.)



## Let's Recall

- A globe is a small model of the Earth with oceans, seas and continents shown on a smaller scale.
- The Earth is spherical in shape.
- The astronomers have found that the axis of our Earth is tilted  $23\frac{1}{2}^\circ$  from the perpendicular position.
- The top end or the northernmost point of the axis is known as North Pole and the bottom end or the southernmost point is called the South Pole.
- The Earth has two movements—rotation and revolution.
- Rotation refers to the spinning of the Earth on its own axis.
- The movement of the Earth around the sun on a fixed path is called revolution.
- Different seasons are caused because of the inclined axis and the revolution of the Earth around the sun.
- The positions of the Earth on 21 March and 23 September, when the days and nights are of equal length, are known as equinoxes.
- The revolution of the Earth around the sun and the revolution of the moon around the Earth cause eclipses.

## Exercises

### A Tick the correct answer.

1. What does the part of the Earth that gets sunlight experience?  
☐ Dusk                      ☐ Night                      ☐ Day
2. Around which heavenly body does our Earth revolve?  
☐ Solar system                      ☐ Sun                      ☐ Moon
3. Earth rotates in which direction?  
☐ West to east                      ☐ North to south                      ☐ East to west
4. During which eclipse the moon may turn reddish?  
☐ Solar eclipse                      ☐ Lunar eclipse                      ☐ Total eclipse

### B Fill in the blanks.

1. Astronomers have found that the \_\_\_\_\_ of our Earth does not stand upright.
2. The Earth moves around the sun on a fixed elliptical path known as \_\_\_\_\_.
3. 21 March and 23 September are known as \_\_\_\_\_.
4. \_\_\_\_\_ happen when the moon, the sun and the Earth fall in the same line.

### C Write short notes on the following.

1. Equinoxes
2. Summer Solstice and Winter Solstice

### D Answer the following questions.

1. Describe the two movements of the Earth.
2. What does inclination of the Earth's axis mean?
3. What are the reasons for changes in the seasons around the year?
4. Explain the solar and lunar eclipses with the help of diagrams.

### E Give reasons.

1. A leap year has 366 days.
2. You should not watch the solar eclipse with naked eyes.
3. You can watch the lunar eclipse with naked eyes.

### 3

## Structure of the Earth

### Let's Learn About

- Continental Drift Theory
- Interior of the Earth
- Exterior of the Earth
- Changes on the Surface of the Earth

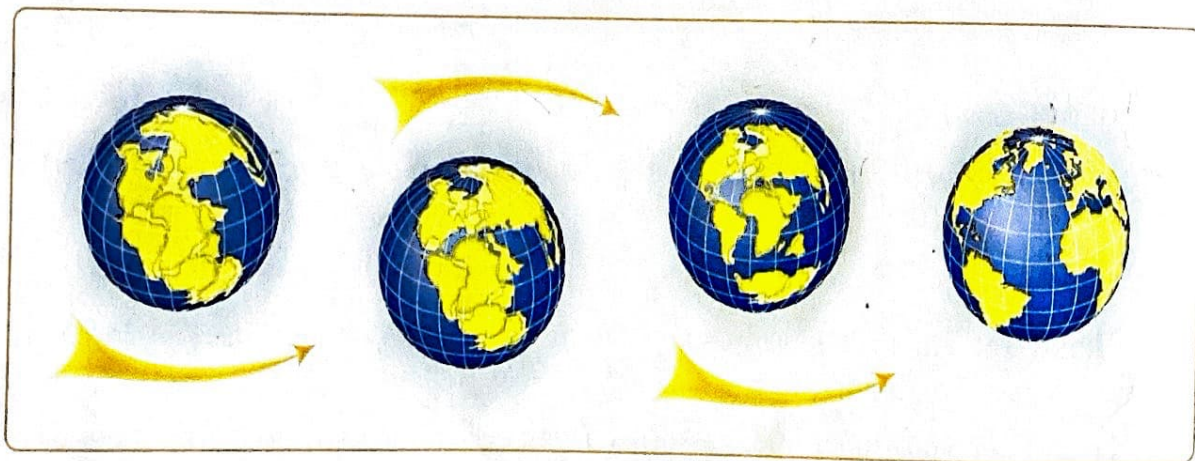
### Warm Up

Are you familiar with the natural happenings like earthquake, cyclone, flood or a drought? Talk about any one of them in class.

Our Earth is a heavenly body in the universe. According to the 'Big Bang Theory' the Earth was made up of molten material that exploded from a hot ball of fire. Gradually, the molten material cooled and solidified, forming a thin hard surface layer called the crust, on which we live. The interior of the Earth is still very hot.

### Continental Drift Theory

The present structure of the Earth was different 200 million years ago. There were no continents and oceans. All the landmasses together formed a supercontinent known as Pangaea. There was only one water body known as Panthalassa. Due to internal movements of the Earth, the landmass started to divide into pieces like a jigsaw puzzle. These pieces were known as plates and they floated on the liquid rock matter. Gradually, these pieces began to move away from each other forming continents. This theory was stated by Alfred Wegener, a German scientist, and was known as the Continental Drift Theory. Later, another theory proved that not only the continents but also the ocean beds move. This theory is known as the Plate Tectonic Theory.



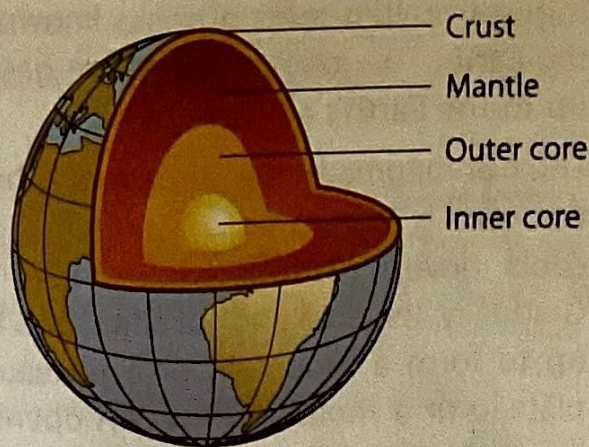
Continental Drift

## Interior of the Earth

The Earth's interior is made up of three layers, namely crust, mantle and core. Photographs of the Earth taken from space show continents and oceans of the Earth. Below the oceans is the solid floor known as the ocean floor. The continents and the ocean floor together form the crust of the Earth.

Mantle is the second layer of the Earth. It is made of rocks rich in iron and other metals.

Core is the centre of the Earth and is made of two heavy metals, nickel and iron. Hence this layer is also known as NiFe (Ni and Fe are the symbols for nickel and iron respectively).



Interior of the Earth

## Exterior of the Earth

The surface of the Earth is made of 29% land and 71% water. The large landmasses are known as continents. Asia, Africa, North America, South America, Europe, Antarctica and Australia are the seven continents of the world.

Other than these, there are small islands scattered across the oceans. The continents are further divided into countries. As per area, Russia is the largest and Vatican City is the smallest country. India is the seventh largest country in the world.

The very deep and large water bodies on the Earth are divided into five oceans, namely the Pacific, the Atlantic, the Indian, the Arctic and the Southern (or Antarctic) Oceans. Besides oceans, there are many other smaller water bodies like the seas, rivers, lakes, ponds, gulfs, bays and streams. All these water bodies are useful to us in many ways. They are home to many water animals and are also a source of water transport, salt and electricity.

## Changes on the Surface of the Earth

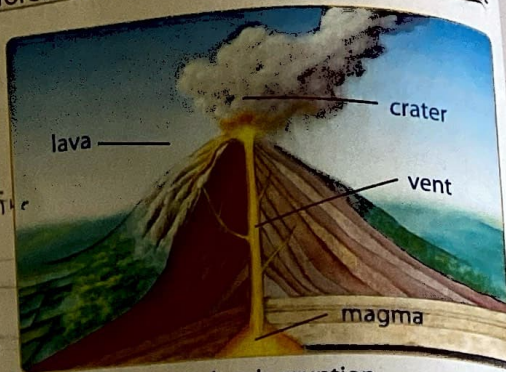
There are constant changes both inside and outside the surface of the Earth. They are known as rapid and slow changes. Slow changes are caused by rain, wind, sun, rivers, seas and glaciers. They take place outside the surface of the Earth and are responsible for changing the Earth's outer surface. Rapid changes are the movements in the interior of the Earth. They cause things like volcanic eruptions and earthquakes.

## Volcanoes

As we go deeper inside the Earth, the temperature keeps on increasing. It is very high at the core. The high temperature melts the rocks in the interior of the Earth and

forms a molten mass of rocks known as **magma**. Some gases are also present inside the Earth. The magma and these gases are forced out through a passage called **vent** up to the Earth's surface.

The hot magma that is thrown out through the vent is known as **lava**. When the lava cools down, it turns into a hard rock. Gradually, the cool and solidified lava builds up to form a mountain or a plateau. (This plateau or a mountain with an opening at its centre through which the molten lava, gases and rock pieces are thrown out is called a **volcano**). The mouth of the volcano is funnel-shaped and is called a **crater**.



Volcanic eruption

#### Learn More

- Mt Cotopaxi in South America is one of the highest active volcanoes in the world.
- The Hawaiian Islands are formed due to volcanic activity.

#### Do and Learn

Make a model of a volcano using clay or play dough and label its parts.

### Earthquakes

(The magma and the gases in the Earth's interior move and push the land upward, making the Earth tremble or vibrate. This violent trembling is known as an **earthquake**). It occurs without any warning and causes a lot of destruction to life and property. An example of this is the earthquake at Bhuj in Gujarat that took place on 26 January 2001. This earthquake claimed the lives of more than 20,000 people and left over 15 lakh people injured. About 8 lakh houses were also destroyed in this disaster.



Gujarat earthquake (26 January, 2001)

The place where an earthquake begins, below the Earth's surface, is called **focus**. The point on the Earth's surface, just above the focus, is called **epicenter**. This is the place where the effect of the earthquake is the maximum.

**Seismology** is the study of earthquakes. Earthquakes are measured and recorded by an instrument called **seismometer** and their intensity is measured with the Richter scale.

Certain places on the Earth experience earthquakes. For example, the Himalayan ranges in northern India are prone to earthquakes.

Given below are some precautions we can take:

- Come out of an enclosed area and move to an open space.
- Shift to the corner of the room, away from windows, glass doors, and heavy furniture that may fall on you.
- Preferably use stairs and not lifts during an earthquake.
- If travelling, come out of the vehicle and move to an open space in case of an earthquake.
- Take care not to stand below trees, power lines, etc., on the road.

### Tsunami

A **tsunami** (pronounced soo-nah-tee) is a series of underwater waves, created by the displacement of a water body by an earthquake, volcanic eruption or any other reason. Tsunami waves are different from the waves created by wind and are quite small compared to tsunami waves which are huge and travel very quickly causing huge destruction when they hit.

Tsunami is a Japanese word which means 'harbour wave'. The 2004 tsunami was one of the deadliest in history.

### Floods

Floods usually occur due to heavy rainfall. As a result, water bodies like rivers overflow and the water spreads over the surrounding areas. Even in areas where rainfall is the main reason for flooding, it can happen because of the following reasons:

- Landslides which result in the blocking of channels

Certain places on the Earth experience earthquakes more often than the others. For example, the Himalayan ranges in northern India, Japan and the Philippines are more prone to earthquakes.

Given below are some precautions we can take during an earthquake.

- Come out of an enclosed area and move to an open place.
- Shift to the corner of the room, away from windows, fans or any other things that may fall on you.
- Preferably use stairs and not lifts during or immediately after an earthquake.
- If travelling, come out of the vehicle immediately in case of an earthquake.
- Take care not to stand below trees, signboards, etc., on the road.

#### Class Discussion

Discuss why we should not stand below trees or signboards during an earthquake.

## Tsunami

A **tsunami** (pronounced soo-nah-mee) is a series of underwater waves, created in an ocean or a water body by an earthquake, landslide, volcanic eruption or any other reason. Tsunami waves are different from the waves we usually find on the coasts. Coastal waves are made by wind and are quite small compared with tsunami waves which are huge and can travel very quickly causing huge destruction when they hit.



Destruction caused by tsunami waves

#### Learn More

Tsunami is a Japanese word which means 'great harbour wave'. The December 2004 Asian tsunami was one of the deadliest in recorded history with a death toll of nearly 3,00,000.

## Floods

**Floods** usually occur due to heavy rainfall. As a result, water bodies like rivers and lakes overflow and the water spreads to their banks and surrounding areas. Even though heavy rainfall is the main reason of floods, it can also happen because of the following reasons:

- Landslides which result in blocking of river channels



Flood

- Damage to a dam built across a river
- Cyclones and tsunamis in coastal areas
- Strong winds like typhoons and storms in the sea
- Melting of ice on mountains

Being a peninsular country which is surrounded by the Arabian Sea, the Indian Ocean and the Bay of Bengal, floods are quite common in India. The plains of northern India suffer from floods frequently.

Floods cause a lot of damage and destruction to the life of people, property and crops. The after-effects of floods are also destructive. Floods can spread dangerous diseases like cholera, malaria and dengue. They can cause waterlogging also.

The only way to prevent floods is to grow trees. Deforestation is the main reason why floods occur easily. The roots of trees absorb water and can prevent floods from spreading fast. Other than this, we can prepare ourselves to face floods in the following ways.

- Keep stocks of food and freshwater.
- Listen to radio or watch television for warnings and advises.
- Keep a first-aid kit in case of any emergency.
- Keep torches, matchboxes and candles handy.
- If water enters inside your house, switch off all electronic devices.

## Droughts

**Drought** is a period of dry weather which happens due to lack of rainfall. All the water reservoirs like rivers, lakes, ponds and wells dry up and there is an extreme shortage of water for agriculture and domestic purpose.

**Famine** is the major consequence of a drought. Farmers are the worst-affected people. They are not able to grow crops as there is little water. This leads to shortage of food and it becomes a famine. Apart from humans, animals and plants are also affected because of famines, as water is an integral part of any living thing. Forest fires occur mostly during this time.

There are a number of ways by which droughts can be prevented. They are as follows.

- By growing more trees
- By growing drought-resistant crops such as millets, sorghum and maize
- By constructing reservoirs to hold emergency water supplies
- By developing rainwater harvesting systems



Drought

## Cyclones

Cyclones are high speed winds which move in circles. They are always accompanied by heavy showers.

Cyclones can be very disastrous and can cause heavy damage to life and property. The most affected areas are the coastal regions. In India, the eastern coast is more affected than the western coast. The state of Odisha has suffered the most due to cyclones. A very powerful cyclone which hit Odisha on 29 October 1999, resulted in the deaths of thousands of people. A lot more became homeless.

We can prepare ourselves when there is a cyclone attack in the following ways.

- Listen to the radio or television for warnings and act accordingly.
- Move to the strongest part of your home when a cyclone hits.
- Stay away from windows.
- Store food in waterproof bags.
- Keep a first-aid kit ready.
- Don't go out of the house until the cyclone stops.



Cyclone

## Let's Recall

- According to the 'Big Bang Theory', the Earth was made up of molten material that exploded from a hot ball of fire.
- The surface of the Earth is made of 29% land and 71% water. The large landmasses are known as continents. The very deep and large water bodies are called oceans.
- A mountain or a plateau with an opening through which the molten lava, gases, and rock pieces are thrown out is called a volcano.
- The violent trembling of the Earth's crust is known as an earthquake.
- A tsunami is a series of underwater waves, created in an ocean or a water body by an earthquake, landslide, volcanic eruption or any other reason.
- Floods usually occur due to heavy rainfall resulting in overflow of water from water bodies.
- Drought is a period of dry weather which happens due to lack of rainfall.
- Cyclones are high speed winds which move in circles and are accompanied by heavy showers.

## Exercises

### A Tick the correct answer.

- Who among the following gave the Continental Drift Theory?  
☐ Alfred Nobel      ☐ Alfred Wegener      ☐ Albert Einstein
- Which is the top layer of the Earth?  
☐ Core      ☐ Mantle      ☐ Crust
- What is the hot magma that comes out through the vent of the volcano known as?  
☐ Magma      ☐ Lava      ☐ Crater
- At which place is the effect of an earthquake the maximum?  
☐ Focus      ☐ Bottom      ☐ Epicenter

### B Fill in the blanks.

- The surface of the Earth is made of 29% \_\_\_\_\_ and 71% \_\_\_\_\_.
- Nearly 200 million years ago, all the landmasses formed a supercontinent known as \_\_\_\_\_.
- The metals nickel and iron make up the \_\_\_\_\_ of the Earth.
- A \_\_\_\_\_ is a series of underwater waves.

### C Answer the following questions.

- H/W {
- What is Continental Drift Theory?
  - Write a note on the exterior of the Earth.
  - Compare slow changes and rapid changes. Give their examples.
  - What is a volcano?
  - How does an earthquake occur?

### D Give reasons.

- H/W {
- The core of the Earth is known as NiFe.
  - Famine is a major consequence of droughts.
  - Floods are quite common in India.